Ruby Shrestha

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Research Interests

Reliable Deep Generative Models; Causal Generative Modeling; Controllable Generation; Robustness to Distribution Shift; Trustworthy ML

Education

PhD, Computing and Information Sciences 08/2022–Present
Rochester Institute of Technology (RIT), Rochester, NY, USA GPA: 4.00/4.00
Advisor: Dr. Linwei Wang
Relevant Courses: Quantitative Foundations, Deep Learning, Statistical Machine Learning, Non-convex Optimization, Data Science & Analytics, Visual Analytics

B.Sc., Computer Science and Information Technology 10/2013–04/2018
Tribhuvan University, Nepal 84.14%

Technical Skills

Programming:	Python, SQL
Machine Learning:	PyTorch, Scikit-learn, NumPy, Pandas
Deep Learning / Research:	Generative Models (Diffusion Models, VAEs, GANs), Causal Modeling, Interventional and Counterfactual Generation
Experiment Tracking:	Weights & Biases, MLflow
Data Systems:	MySQL, Elasticsearch
Tools / Platforms:	Git, Linux, Mac OS, Windows

Research Experience

Graduate Research Assistant 2022–Present
Computational Biomedicine Lab, Rochester Institute of Technology, Rochester, NY, USA

- Conducted a systematic study of how attribute correlations in training data impair generative models' ability to produce underrepresented attribute combinations, and the extent to which existing mitigation strategies address this impact (journal, under review).
- Developed a deep generative modeling framework that improves generation of underrepresented attribute combinations, validated across VAEs, GANs, and diffusion models (first-author, ICLR 2026).
- Developing causal generative model for controllable and counterfactual image generation that is robust to varying strengths of causal relationships between attributes.

Independent Research Study 05/2018–08/2018
Deerwalk Institute of Technology, Tribhuvan University, Nepal

- Proposed a hybrid path-finding approach combining irregularities-tolerant dead-end filling with A* search for convoluted maze paths, leading to a journal article (DJCSIT, 2019).

Research Assistant 03/2017–05/2017
Deerwalk Institute of Technology, Tribhuvan University, Nepal

- Investigated student perceptions of blended learning in higher education, with findings presented at IT4D 2017.

Publications

ProReGen: Progressive Residual Generation under Attribute Correlations
R. Shrestha, A. Gopi, C. Meisenzahl, B. Lekhak, L. Wang
International Conference on Learning Representations (ICLR), 2026. Conference Article.

DATS: Difficulty-Aware Task Sampler for Meta-Learning Physics-Informed Neural Networks
M. Toloubidokhti, Y. Ye, R. Missel, X. Jiang, N. Kumar, **R. Shrestha**, L. Wang
International Conference on Learning Representations (ICLR), 2024. Conference Article.

Distributionally Robust Optimization and Invariant Representation Learning for Addressing Subgroup Underrepresentation: Mechanisms and Limitations
N. Kumar*, **R. Shrestha***, Z. Li, L. Wang *Equal contribution
Fairness of AI in Medical Imaging (FAIMI @ MICCAI), 2023. Workshop Article.

Maze Solving with Anomaly Tolerant Dead End Filling and A*: Implementation and Analysis
R. Shrestha
Deerwalk Journal of Computer Science and Information Technology (DJCSIT), 2019. Journal Article.

Industry Experience

Software Consultant 04/2021–07/2022
Saugal Trading Company Pvt. Ltd., Nepal

Senior Software Engineer, Development Manager 01/2021–03/2021
Software Engineer, Team Lead 04/2019–01/2021
Software Engineer 05/2018–03/2019
Software Engineering Intern 09/2015–04/2018
Deerwalk Inc. (now a Cedar Gate Technologies Company)

- Developed and led projects to optimize Elasticsearch queries and build healthcare data extraction frameworks for industry-standard descriptive, prescriptive, and trend-based healthcare reports and data visualizations.

Honors & Awards

- Dr. Russell & Melissa Bessette Award for Doctoral Student Excellence, RIT (2023)
- National Female Education Award, Government of Nepal (2019)
- Winner, International Girls in ICT Day Hackathon (2016)
- Outstanding Cambridge Learners Award – Top in Country (2013)

Conference Presentations

ProReGen: Progressive Residual Generation under Attribute Correlations 04/2026
Poster, The Fourteenth International Conference on Learning Representations (ICLR)

Maze Solving with Anomaly Tolerant Dead End Filling and A*: Implementation and Analysis 08/2018
Oral, The Fifth International Conference on Information Technology for Development (IT4D)

Blended Learning in DWIT College: From Learners Perspective 07/2017
Oral, The Fourth International Conference on Information Technology for Development (IT4D)

Teaching Experience

Guest Lecturer 11/2025

CISC-865 Deep Learning, Rochester Institute of Technology, Rochester, NY, USA

- Lectured on "Robustness of Deep Learning Models to Observed Attribute Correlations: Exploring Impact and Potential Solutions."

Workshop Instructor 01/2019–03/2019

Deerwalk Institute of Technology, Tribhuvan University, Nepal

- Taught a workshop on fundamental concepts of data management and analysis with Elasticsearch.

Guest Trainer 07/2018

Local Development Training Academy, Nepal

- Delivered a three-day hands-on training program on cloud services, content management systems, and project management tools.